Anikait Singh

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Research Interests

My research develops algorithms for self-improving and creative foundation models, with the goal of enabling progress on difficult, open-ended problems such as scientific research. These problems require more than imitating demonstrations or solving well-specified tasks: models must decide what is worth trying, synthesize partial evidence, discover useful abstractions, and refine ideas whose value is revealed only through downstream validation. I study how feedback from a model's own attempts, natural-language critiques, and inference-time computation can be transformed into learning signals that drive exploration and improvement. The aim is to build models that can generate, evaluate, and refine novel insights, learning strategies that generalize across domains.

Education

Stanford University, Palo Alto, CA — Sep 2023 – Present
Ph.D. in Computer Science — Advisors: Prof. Chelsea Finn and Aviral Kumar

University of California, Berkeley, Berkeley, CA — Aug 2019 – May 2023
B.A. in Computer Science (Highest Distinction) — GPA: 3.986; Technical GPA: 4.0
Selected coursework: Deep Learning, Reinforcement Learning, Convex Optimization, NLP, and Info Theory.

Experience

Meta Superintelligence Labs, Paris, France — Jun 2026 – October 2026
Research Scientist Intern.

Microsoft Research, New York, NY — Jun 2025 – Sep 2025
Research Scientist Intern.

Synthlabs AI, San Francisco, CA — Jan 2025 – March 2025
Research Collaboration.

Toyota Research Institute (TRI), Los Altos, CA — Jun 2024 – Sep 2024
Research Scientist Intern.

Google DeepMind Robotics, Mountain View, CA — Apr 2023 – Feb 2024
Student Researcher.

X, the moonshot factory, Mountain View, CA — Dec 2022 – Apr 2023
AI Resident.

Selected Publications

(* equal contribution; selection)

- **Learning to Discover Abstractions for LLM Reasoning**. Y. Qu*, **A. Singh***, Y. Lee*, A. Setlur, R. Salakhutdinov, C. Finn, A. Kumar. ICLR 2026.
- **Meta Rubrics: Learning to Discover Rubrics at Deployment for Efficient Task Adaptation**. **A. Singh***, A. Kumar, C. Finn, R. Munos. Under Submission to ICLR 2027.
- **Librarian Agents: Anticipatory Filesystem Construction via Reinforcement Learning**. **A. Singh***, T. Zhang*, Y. Lee, A. Kumar, C. Finn. CoLM 2026 LLA Workshop. Under Submission to ICLR 2027.
- **GIANTS: Generative Insight Anticipation from Scientific Literature**. J. He-Yueya*, **A. Singh***, G. Gao, M. Y. Li, S. Yang, C. Finn, E. Brunskill, N. D. Goodman. CoLM 2026 LM4Sci Workshop (Oral). Under Submission to ICLR 2027.

- **Few-Shot Preference Optimization of Synthetic Preferences Elicits Effective Personalization to Real Users**. **A. Singh***, S. Hsu*, E. Mitchell, T. Hashimoto, A. Sharma, C. Finn. ICLR 2026.
- **Cognitive Behaviors that Enable Self-Improving Reasoners**. K. Gandhi, A. Chakravarthy, **A. Singh**, N. Lile, N. D. Goodman. CoLM 2025 (Oral).
- **Improving Test-Time Search in LLMs with Backtracking Against In-Context Value Verifiers**. **A. Singh**, K. Arora, S. Keh, T. Hashimoto, C. Finn, A. Kumar. ICLR 2025 Workshop (Best Paper Runner-Up).
- **Personalized Preference Fine-tuning of Diffusion Models**. M. Dang*, **A. Singh***, L. Zhou, S. Ermon, J. Song. CVPR 2025.
- **Towards System 2 Reasoning in LLMs: Learning How to Think With Meta Chain-of-Thought**. V. Xiang, C. Snell, K. Gandhi, A. Albalak, **A. Singh**, C. Blagden, D. Phung, R. Rafailov, N. Lile, D. Mahan, L. Castricato, J.-P. Franken, N. Haber, C. Finn. Position Paper, 2024.
- **Preference Fine-Tuning of LLMs Should Leverage Suboptimal, On-Policy Data**. **A. Singh***, F. Tajwar*, A. Sharma, R. Rafailov, J. Schneider, T. Xie, S. Ermon, C. Finn, A. Kumar. ICML 2024.
- **Robotic Offline RL from Internet Videos via Value-Function Pre-Training**. C. Bhateja*, D. Guo*, D. Ghosh*, **A. Singh**, M. Tomar, Q. Vuong, Y. Chebotar, S. Levine, A. Kumar. ICRA 2024.
- **Offline RL With Realistic Datasets: Heteroskedasticity and Support Constraints**. **A. Singh***, A. Kumar*, Q. Vuong, Y. Chebotar, S. Levine. NeurIPS 2023.
- **Cal-QL: Calibrated Offline RL Pre-Training for Efficient Online Fine-Tuning**. M. Nakamoto*, Y. Zhai*, **A. Singh**, M. S. Mark, Y. Ma, C. Finn, A. Kumar, S. Levine. NeurIPS 2023.
- **Pre-Training for Robots: Offline RL Enables Learning from a Handful of Trials**. A. Kumar*, **A. Singh***, F. Ebert*, M. Nakamoto, Y. Yang, C. Finn, S. Levine. RSS 2023.
- **RT-2: Vision-Language-Action Models Transfer Web Knowledge to Robotic Control**. Google DeepMind Robotics. ICRA 2023.
- **When Should We Prefer Offline Reinforcement Learning Over Behavioral Cloning?**. A. Kumar, J. Hong, **A. Singh**, S. Levine. ICLR 2022.
- **A Workflow for Offline Model-Free Robotic Reinforcement Learning**. A. Kumar*, **A. Singh***, S. Tian, C. Finn, S. Levine. CoRL 2021 (Oral).

Teaching

Mentoring Program Organizer, Stanford CS Undergraduate Mentoring Program (2024-Present).
Organizer, Stanford NLP Seminar (2025-2026).
Program Coordinator & Mentor, Deep Learning Portal (2024).
Graduate Student Instructor, CS224r (Spring 2025/2026, Project Lead).

Honors

NSF Graduate Research Fellowship (2023–2028).
Gemini Academic Program Award (2025)
Tinker & Modal Academic Research Grants (2025/2026)

Invited Talks and Service

Invited Talks at Engram, Salesforce Future AI Forum, Jump Trading, and ServiceNow, BeNeRL
Reviewer (w/honors) for ICLR, NeurIPS, ICML, CoRL, ICRA (RAL), AAAI, and CoLM.